Fisher Information

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Introduction

Fisher information is a numerical summary of how much information a data sample carries about an unknown parameter. When the likelihood is sharply peaked, small changes in the parameter produce big changes in the data's probability, so the data tell us a lot about the parameter – the information is high. When the likelihood is flat, the data are nearly indifferent to the parameter, so information is low. This intuitive idea turns into a precise quantity that drives the asymptotic theory of maximum likelihood.

Prerequisites

The reader should be comfortable with calculus, with the likelihood function, and with the notion of taking an expectation.

Theory

For a parametric density $f(x \mid \theta)$, the **score function** is the gradient of the log-likelihood for a single observation:

$$U(\theta; x) = \frac{\partial}{\partial \theta} \log f(x \mid \theta).$$

Its expected value under the true parameter is zero; its variance is the **Fisher information**:

$$I(\theta) = \mathrm{Var}[U(\theta; X)] = E\left[\left(\frac{\partial \log f}{\partial \theta}\right)^2\right].$$

Under sufficient regularity, this equals the expected negative curvature of the log-likelihood:

$$I(\theta) = -E\left[\frac{\partial^2 \log f}{\partial \theta^2}\right].$$

For n iid observations, $I_n(\theta) = nI(\theta)$ – information is additive across independent observations. The **observed information** replaces the expectation with the empirical curvature at the MLE:

$$\hat{\mathcal{I}}(\hat{\theta}) = -\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^\top}\bigg|_{\theta=\hat{\theta}}.$$

Its inverse is the asymptotic covariance of the MLE, and its inverse diagonal is the squared standard-error estimates reported by every maximum-likelihood fit.

Fisher information is the cornerstone of the Cramer-Rao lower bound, the asymptotic efficiency of the MLE, and optimal experimental design (where the design maximises the determinant of the information matrix).

Assumptions

The classical definition and the curvature-based identity require smoothness of the log-density in θ and the ability to interchange differentiation and integration. Non-regular problems (boundary parameters, truncation points, non-identified parameters) require specialised information quantities.

R Implementation

```r
set.seed(2026)

n <- 200
x <- rnorm(n, mean = 5, sd = 2)

nll <- function(par) {
  mu <- par[1]; sigma <- par[2]
  if (sigma <= 0) return(Inf)
  -sum(dnorm(x, mean = mu, sd = sigma, log = TRUE))
}

fit <- optim(c(0, 1), nll, hessian = TRUE, method = "L-BFGS-B",
             lower = c(-Inf, 1e-6))
obs_info <- fit$hessian
obs_info

exp_info <- matrix(c(n / fit$par[2]^2, 0,
                     0, 2 * n / fit$par[2]^2), 2, 2)
exp_info

se <- sqrt(diag(solve(obs_info)))
se
```

We fit a normal model by ML, extract the observed information from the Hessian, compute the expected information from its closed form, and recover the standard errors by inverting.

Output & Results

```
          [,1]       [,2]
[1,] 48.86      -0.07
[2,] -0.07      97.74

          [,1]       [,2]
[1,] 48.93       0.00
[2,]  0.00      97.87

[1] 0.143 0.101
```

Observed and expected information agree to within simulation noise. The inverse-information standard errors match what R's `summary.lm()` or `summary.glm()` would report for these parameters.

Interpretation

Information is what turns data into precision. Reporting the standard errors of a maximum-likelihood fit amounts to reporting the inverse-square-root diagonal of the information matrix. When planning a study, maximising information per unit of cost (D-optimality, A-optimality, I-optimality in optimal design) is the mathematical expression of "get as much precision as possible from the resources available".

Practical Tips

- For standard errors, use observed information $\hat{\mathcal{I}}(\hat{\theta})$; it is better-behaved than expected information in small samples.
- When the log-likelihood is flat (weakly identified parameters), the information matrix is near-singular and SEs are huge. This is usually a sign of a specification or identifiability issue, not an artefact.
- In multivariate models, the off-diagonal entries of the information matrix quantify the coupling between parameters; a large off-diagonal means the joint uncertainty is not captured by marginal SEs alone.
- Regularity is the fine print: in problems where standard MLE theory fails, check whether a modified or restricted information concept applies.
- The Fisher information of an estimator is not the same as the information in the data; be careful not to conflate the two quantities.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr